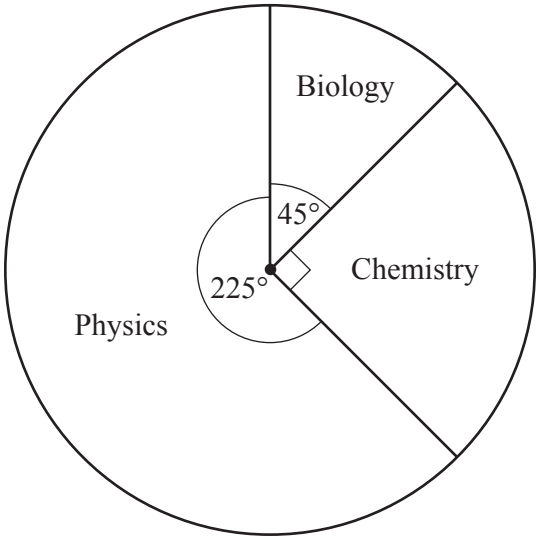
0580_s23_qp_33_Q1

1 **(a)** Claudia asks some students to choose their favourite science from biology, chemistry and physics. The pie chart shows the results.



(i) Find the percentage of students who choose chemistry.

..... % [1]

(ii) Find the fraction of students who choose physics.
Give your answer in its simplest form.

..... [2]

(iii) For the number of students choosing each subject, find the ratio biology : chemistry : physics.
Give your answer in its simplest form.

..... : ..... : ..... [2]

(iv) Marcus says:

'I do not know how many people choose chemistry, but I do know it is an even number.'

Explain how Marcus knows this.

..... [1]

(v) Claudia now tells Marcus that 26 students choose chemistry.

Work out how many students choose physics.

..... [2]

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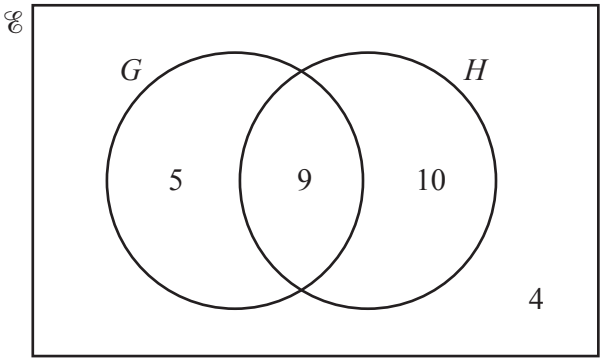
(b) The Venn diagram shows information about the number of students in a class who study geography (G) and history (H).



(i) Work out the number of students in the class.

..... [1]

(ii) Find $n(G)$.

..... [1]

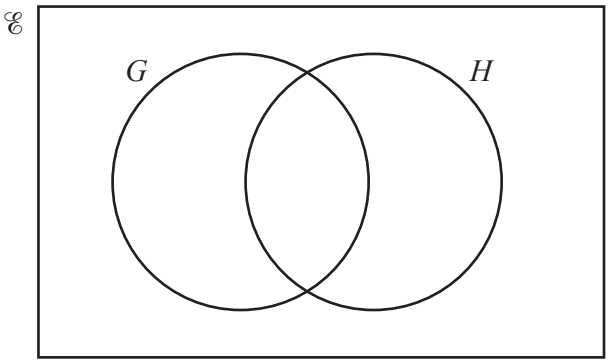
(iii) One of the students is chosen at random.

Find the probability that this student studies geography and history.

..... [1]

(iv) One of the students who studies geography and history stops studying history.

Complete this Venn diagram to show this change.



[1]